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STAFF REPORT FOR THE 2018 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Cambodia Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Low*
Overall risk of debt distress	*Low*
Granularity in the risk rating	*Not Applicable*
Application of judgment	No

The Debt Sustainability Analysis (DSA) using the revised joint IMF/WB Debt Sustainability Framework for Low Income Countries (LIC-DSF) shows that Cambodia remains at low risk of external debt distress. All debt burden indicators are projected to remain under their indicative thresholds under the baseline and the shock scenarios.[1] While the overall risk of debt distress is low, the analysis indicates that debt sustainability is vulnerable to export and growth shocks, and the materialization of contingent liabilities. Furthermore, the public and publicly guaranteed (PPG) debt-to-GDP ratio is projected to rise by more than 10 percentage points during the next decade due to projected large fiscal deficits in the medium term. These findings reinforce the importance of implementing reforms to increase the economy's resilience to external shocks and encourage export and economic diversification. Efforts to mobilize fiscal revenue, to strengthen the Public Investment Management framework (including for PPPs) and further enhance monitoring of PPP and financial sector risks, as well as introduction of a debt-based fiscal anchor, are necessary to ensure debt sustainability in the medium term. .

[1] Cambodia's Composite Indicator (CI) index, which has been calculated based on the October 2018 WEO and 2017 CPIA update indicates that the county's debt-carrying capacity is medium. This classification remains unchanged from the previous DSA framework (grounded in CPIA readings), but thresholds are updated to those of the revised DSF. For the PV of PPG external debt, thresholds are increased from 30 to 40% (GDP) and from 100 to 180% (exports); external debt service-to exports and to revenue are unchanged at 15% and 18% respectively.

PUBLIC DEBT COVERAGE

1. The DSA covers the central government debt as well as the debt guaranteed by the central government to state-owned enterprises (SOEs). By law, state and local governments and the central bank do not engage in external borrowing, and SOEs do not contract non-guaranteed external loans. There are no extra budgetary funds and the National Social Security Fund is funded by deposits and does not constitute a liability for the general government.

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

The following contingent liabilities are included in the contingent liability stress test: PPP (6.3 percent of GDP) and financial market (5 percent of GDP). [2]

1	The country's coverage of public debt	The central government, government-guaranteed debt			
		Default		Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	0.0	Not allowed by law.
4	PPP	35	percent of PPP stock	6.3	
5	Financial Market	5	percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)			11.3	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON DEBT

2. Cambodia's external public debt, including arrears, amounted to US$6.7 billion (30.3 percent of GDP) at end-2017. The external debt-to-GDP ratio has remained relatively flat during the last decade, increasing by 0.2 percentage points on average per year. The composition of the debt stock is 70 percent bilateral debt—more than half of it owed to China—with the remaining 30 percent multilateral debt. External debt has been accrued on concessional terms, and the PV of debt-to-GDP ratio is calculated at 23.6 percent for end-2017. The debt stock includes legacy arrears to the Russian Federation and the United States of about 3 percent of GDP.[3] The status of negotiations of these arrears remains unchanged compared to the last DSA and this analysis continues to assume no debt restructuring.

Cambodia: External Public Debt, 2017

	In million of U.S. dollars	In percent of GDP	In percent of external debt
Total	6668.6	30.3	100
Multilateral	1951.0	9.0	29.3
Bilateral	4,717.6	21.3	70.7
of which: China	3,209.6	14.5	48.1

Sources: Cambodia authorities; and IMF estimates.

[2] The PPP stock is estimated using IMF's Investment and Capital Stock Dataset.

[3] Based on Cambodia Public Debt Statistical Bulletin (see Table 13 "Old Debt Under Negotiation"). Data reflects principal amounts, i.e. excluding any accumulated interest. The arrears relate to borrowing prior to 1993.

3. Public domestic debt is currently negligible. The stock as of end-2017 was US$2.8 million. To support financial market development, the government is preparing to issue, for the first time, local-currency government bonds over the next few years.

4. The PPP framework continues to be strengthened while the stock of PPP has increased. With expected diminished access to concessional financing and slow progress in developing domestic debt markets, PPPs have been identified as a way to finance necessary investment projects. The PPP stock grew more than twofold between 2010 and 2015, when it was estimated at 17 percent of GDP. There is no recent estimation of the Cambodian PPP stock. The authorities have established a roadmap to have a full set of PPP mechanisms by 2020, including the legal framework and system for risk assessment. Technical assistance provided by the IMF, the World Bank and ADB has focused on developing the Ministry of Economy and Finance's capacity to analyze fiscal risks from PPPs, integrating PPP risks in the assessment of fiscal sustainability and strengthening the legal, regulatory and institutional framework of PPP management. A central PPP unit has been established under the Ministry and the authorities adopted an annual ceiling at 4 percent of GDP. A new PPP law is being drafted with the support of the ADB.

5. Private external debt. The stock of private external debt in Cambodia is not published by the authorities.[4] Risks emerging from excessive external borrowing by the private sector could increase the government's exposure to contingent liabilities. For this analysis, the default magnitude of the contingent liability stress test is used, amounting to 11.3 percent of GDP.

BACKGROUND ON MACRO FORECASTS

6. Growth and inflation. Economic activity is expected to remain strong, driven by construction, tourism and exports. GDP growth is projected at 7½ percent in 2018 and expected to decline towards its potential of 6 percent over the medium term. Inflation was stable at 2.9 percent in 2017 and is expected to decrease slightly to 2.5 percent in 2018, in part owing to measures to contain fuel and food price increases.

7. External sector. The current account deficit was 8 percent of GDP in 2017. On the back of strong FDI inflows and robust remittances, foreign reserves reached US$8.8 billion (5.1 months of imports). The current account deficit is expected to widen in 2018 due increased imports of construction material and to narrow over the medium as economic activity becomes driven more by exports and tourism. In line with authorities' projections, external debt disbursements are expected to average 3.9 percent of GDP annually over 2018–23. External borrowing is projected to remain largely concessional over the next decade and external debt is projected to reach 34.1 percent of GDP by 2023 and to stabilize at 38 percent of GDP in 2028.

8. Fiscal. The fiscal deficit narrowed to 1.1 percent of GDP in 2017, well below the budget deficit target of 3.9 percent, due to significant tax revenue overperformance. Consequently, the level of government deposits rose to 12.6 percent of GDP by end-2017. The fiscal deficit is projected to increase to about 2.2 percent of GDP in 2018 due a rising public sector wage bill, higher capital and social spending.

[4] According to CEIC data, sourced from the NBC, Cambodia's total external debt amounted to US$11.0 billion in 2017. Although the database does not publish the public/private breakdown, private debt can be estimated at about US$4.3 billion (19.4 percent of GDP) by deducting PPG external debt.

The fiscal deficit is projected to widen further over the medium-term, to 3.7 percent of GDP by 2023, as revenue growth moderates—absent of tax policy reforms–while capital and social expenditures are kept at a higher level guided by the government's National Social Protection Policy Framework (2016–2025).

9. Domestic debt. As Cambodia's financial sector continues to develop and access to concessional financing becomes more limited, it is expected that the government will start issuing domestic government bonds that will provide additional fiscal financing. Hence, this analysis assumes that the government will issue long-term domestic bonds starting in 2021 under technical assistance support by the World Bank. The annual issuance of domestic bonds is projected to increase gradually, from 0.4 percent of GDP in 2021 to 1 percent in 2038. Domestic PPG debt is expected to increase from 0 to 4.1 percent of GDP by 2038.

10. Realism of the baseline (Figure 4). Cross-country experience suggests that the baseline fiscal adjustment is feasible. The maximum adjustment over the next 3-year period is a 1.6 percentage point increase in the primary deficit-to-GDP ratio. Given the forecasted fiscal adjustment, the projected growth path is in line with a fiscal multiplier of 0.2. Finally, the contribution of public capital to GDP growth in the baseline scenario is in line with historical values.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

11. Country classification. The revised LIC_DSF determines the debt sustainability thresholds by calculating a *composite indicator* (CI). In the previous DSF, debt-carrying capacity was wholly determined by the World Bank CPIA score. The CI captures the impact of the different factors through a weighted average of the country's real GDP growth, remittances, international reserves, world growth and the CPIA score.[5] The calculation of the CI is based on 10-year averages of the variables, across 5 years of historical data and 5 years of projections. For Cambodia, the CI score based on the 2018 October WEO data, corresponds to a *medium* rating. Therefore, while the previous rating was *strong* (using the 2018 April WEO data), the final classification is kept at *medium*. [6] The corresponding thresholds are noted in the table below. PV of debt-to-exports threshold was increased compared to the previous DSF, from 150 to

Cambodia: Debt Carrying Capacity and Thresholds

Debt Carrying Capacity		
Final	Classification based on current vintage	Classification based on the previous vintage
Medium	Medium 3.01	Strong 3.06

Applicable thresholds	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	180
GDP	40
Debt service in % of	
Exports	15
Revenue	18
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	55

[5] The details on the methodology can be found in the new LIC-DSF guidance note: https://www.imf.org/en/Publications/Policy-Papers/Issues/2018/02/14/pp122617guidance-note-on-lic-dsf
[6] Country classification are revised if two consecutive signals suggest an upgrade or downgrade.

180 percent. Debt service-to-exports and to-revenue thresholds were lowered respectively from 20 to 15 percent and from 20 to 18 percent.

EXTERNAL AND PUBLIC SUSTAINABILITY

12. The external DSA shows that Cambodia's risk of debt distress is low. Under the baseline scenario, the PV of debt ratios never breach their respective thresholds and are projected to remain flat over the projection period. Moreover, the debt service-to-exports and debt service-to-revenue ratios remain well below the thresholds throughout the projection period, partly due to the concessional nature of most debt (Figure 1).

13. Stress tests show that Cambodia's external debt sustainability is most vulnerable to export shocks. For all four indicators, the export shock is identified as the most extreme one, leading to a rise of the ratios larger than for shocks to real growth, fiscal primary balance, exchange rate depreciation and external flows. Under the export scenario, the PV of debt-to-GDP ratio would increase quickly from 23.3 percent in 2018 to 36 percent in 2020, but would remain under the 40 percent threshold.

14. The overall risk of public debt distress is low. Under the baseline, total PPG debt is projected to rise from 30.3 percent of GDP in 2017 to 41.4 percent in 2038 (Table 2). The PV of total debt-to-GDP ratio is expected to increase to 30.2 percent in 2028, as the share of concessional external debt to total debt decreases, but to remain well below the 55 percent benchmark (Figure 2). Debt-to-revenue and debt service-to-revenue ratios are also expected to increase over the next decade.

15. PPG debt is vulnerable to growth and contingent liabilities shocks. A growth shock (calibrated at 1 standard deviation of historical performance) in 2019-20 would lead to a rise of the PV of total debt-to-GDP ratio to 45 percent in 2028. The realization of contingent liabilities related to PPPs and financial stress could increase the debt service-to-revenue ratio from 5 percent in 2018 to 15 in 2021.

CONCLUSION

16. Cambodia remains at low risk of external and overall debt distress. However, the total PPG debt-to-GDP ratio is expected to rise by more than 10 percentage points during the next decade due to projected increasing fiscal deficits between 2018 and 2023. To preserve public debt sustainability in the medium-term, the government should establish a fiscal anchor by introducing a debt ceiling at 40 percent of GDP and develop a medium-term budgetary framework. The baseline projections and the standard stress tests show increasing risks to the external and public debt outlooks. Stress tests indicate that Cambodia's debt sustainability remains vulnerable to shocks to exports, economic growth and contingent liabilities. This reinforces the importance of preserving macroeconomic stability and diversifying the economy and exports to increase resilience to external shocks, improving spending efficiency and the successful implementation of the revenue mobilization strategy. Further efforts to implement sound PIM and PPP frameworks and strengthen analysis of PPP risks are needed. Finally, the authorities should focus on closing data gaps, in particular regarding external private debt and the PPP stock.

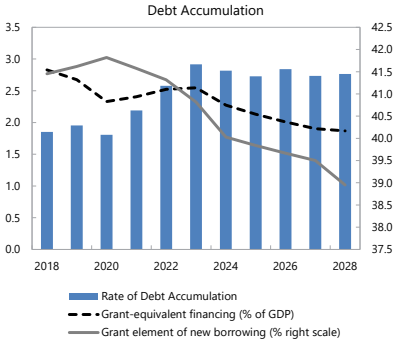
Table 1. Cambodia: Debt Sustainability Framework, Baseline Scenario, 2015–2038

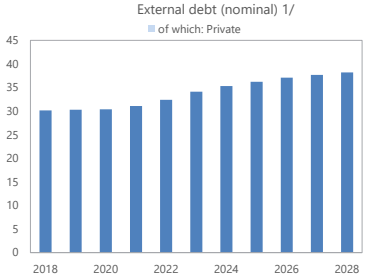
(In percent of GDP, unless otherwise indicated)

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	**31.2**	**29.1**	**30.3**	**30.1**	**30.3**	**30.4**	**31.1**	**32.4**	**34.1**	**38.2**	**37.3**	**30.0**	**33.9**
of which: public and publicly guaranteed (PPG)	**31.2**	**29.1**	**30.3**	**30.1**	**30.3**	**30.4**	**31.1**	**32.4**	**34.1**	**38.2**	**37.3**	**30.0**	**33.9**
Change in external debt	-0.7	-2.0	1.2	-0.1	0.2	0.1	0.7	1.3	1.7	0.5	-0.3		
Identified net debt-creating flows	**-3.4**	**-6.6**	**-6.8**	**-4.6**	**-4.1**	**-3.6**	**-3.9**	**-3.6**	**-3.8**	**-3.1**	**-2.2**	**-5.5**	**-3.7**
Non-interest current account deficit	**8.4**	**8.1**	**7.7**	**9.8**	**9.6**	**9.1**	**8.1**	**7.7**	**7.0**	**5.6**	**1.9**	**8.1**	**7.4**
Deficit in balance of goods and services	12.4	11.2	10.8	11.1	10.9	10.0	8.9	8.6	7.9	6.5	4.4	**12.3**	**8.3**
Exports	73.5	71.4	71.5	74.9	76.8	77.4	77.7	78.0	78.3	74.7	67.0		
Imports	85.9	82.6	82.3	86.0	87.6	87.3	86.6	86.6	86.2	81.2	71.5		
Net current transfers (negative = inflow)	-9.0	-7.8	-7.7	-7.0	-6.9	-6.7	-6.7	-6.8	-6.9	-6.7	-6.5	**-8.7**	**-6.8**
of which: official	-2.9	-2.5	-2.4	-1.8	-1.6	-1.2	-1.1	-1.0	-1.0	-0.8	-0.6		
Other current account flows (negative = net inflow)	5.1	4.6	4.6	5.6	5.6	5.8	5.9	5.9	5.9	5.7	3.9	**4.5**	**5.9**
Net FDI (negative = inflow)	**-9.6**	**-12.0**	**-12.1**	**-12.7**	**-12.1**	**-11.1**	**-10.5**	**-9.9**	**-9.3**	**-7.1**	**-2.6**	**-11.2**	**-9.6**
Endogenous debt dynamics 2/	**-2.1**	**-2.7**	**-2.4**	**-1.7**	**-1.6**	**-1.5**	**-1.5**	**-1.4**	**-1.5**	**-1.6**	**-1.6**		
Contribution from nominal interest rate	0.3	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.5		
Contribution from real GDP growth	-2.1	-1.9	-1.8	-2.0	-1.9	-1.9	-1.8	-1.8	-1.8	-2.1	-2.1		
Contribution from price and exchange rate changes	-0.4	-1.1	-0.9	...	...	...	...	...	...	...	...		
Residual 3/	**2.6**	**4.6**	**7.9**	**4.4**	**4.3**	**3.6**	**4.6**	**4.9**	**5.5**	**3.7**	**1.9**	**5.6**	**4.4**
of which: exceptional financing	-0.1	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**23.6**	**23.3**	**23.2**	**23.0**	**23.2**	**23.8**	**24.8**	**27.5**	**27.2**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**33.0**	**31.1**	**30.2**	**29.7**	**29.9**	**30.6**	**31.6**	**36.8**	**40.6**		
PPG debt service-to-exports ratio	**0.8**	**1.3**	**1.3**	**1.4**	**1.5**	**1.6**	**1.6**	**1.7**	**1.6**	**1.7**	**2.8**		
PPG debt service-to-revenue ratio	**3.3**	**5.1**	**4.8**	**5.3**	**5.5**	**6.2**	**6.2**	**6.4**	**6.3**	**6.0**	**8.4**		
Gross external financing need (Million of U.S. dollars)	-116.6	-595.4	-751.9	-444.1	-359.9	-219.4	-351.3	-289.9	-380.0	-106.5	1532.8		
Key macroeconomic assumptions													
Real GDP growth (in percent)	7.0	6.9	7.0	7.3	7.0	6.7	6.5	6.2	6.0	6.0	6.0	**6.3**	**6.4**
GDP deflator in US dollar terms (change in percent)	1.1	3.7	3.3	1.8	1.8	1.8	1.9	1.9	1.9	2.4	3.0	**3.5**	**2.0**
Effective interest rate (percent) 4/	1.0	1.3	1.2	1.3	1.3	1.3	1.3	1.3	1.2	1.3	1.5	**1.0**	**1.3**
Growth of exports of G&S (US dollar terms, in percent)	10.9	7.6	10.7	14.3	11.6	9.6	8.9	8.6	8.4	7.5	8.0	**13.4**	**9.0**
Growth of imports of G&S (US dollar terms, in percent)	10.1	6.6	10.2	14.0	11.0	8.3	7.6	8.2	7.5	7.4	7.7	**13.5**	**8.4**
Grant element of new public sector borrowing (in percent)	...	...	...	41.5	41.6	41.8	41.6	41.3	40.8	39.0	34.4	**...**	**40.6**
Government revenues (excluding grants, in percent of GDP)	17.7	18.4	20.2	20.4	20.7	20.5	20.5	20.4	20.3	21.1	22.3	**15.1**	**20.6**
Aid flows (in Million of US dollars) 5/	350.4	495.6	420.0	1091.4	1205.6	1198.3	1424.7	1700.5	1934.3	2305.5	2646.6		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.8	2.7	2.3	2.4	2.5	2.5	1.9	1.5	**...**	**2.3**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	61.0	57.5	54.3	50.9	48.3	46.3	41.5	34.8	**...**	**48.6**
Nominal GDP (Million of US dollars)	18,083	20,043	22,147	24,171	26,324	28,609	31,040	33,591	36,285	54,354	127,969		
Nominal dollar GDP growth	8.3	10.8	10.5	9.1	8.9	8.7	8.5	8.2	8.0	8.6	9.2	**10.0**	**8.5**
Memorandum items:													
PV of external debt 7/	...	...	23.6	23.3	23.2	23.0	23.2	23.8	24.8	27.5	27.2		
In percent of exports	...	...	33.0	31.1	30.2	29.7	29.9	30.6	31.6	36.8	40.6		
Total external debt service-to-exports ratio	0.8	1.3	1.3	1.4	1.5	1.6	1.6	1.7	1.6	1.7	2.8		
PV of PPG external debt (in Million of US dollars)			5221.1	5631.3	6103.5	6579.0	7205.7	8005.8	8985.6	14934.2	34836.8		
(PVt-PVt-1)/GDPt-1 (in percent)				1.9	2.0	1.8	2.2	2.6	2.9	2.8	2.3		
Non-interest current account deficit that stabilizes debt ratio	9.1	10.1	6.6	9.9	9.4	9.0	7.4	6.4	5.3	5.0	2.2		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes public sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

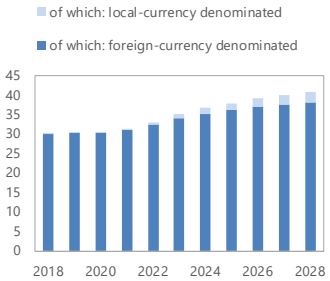
Table 2. Cambodia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–2038

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**31.2**	**29.1**	**30.3**	**30.2**	**30.3**	**30.4**	**31.4**	**33.1**	**35.2**	**40.8**	**41.4**	**30.0**	**35.0**
of which: external debt	**31.2**	**29.1**	**30.3**	**30.1**	**30.3**	**30.4**	**31.1**	**32.4**	**34.1**	**38.2**	**37.3**	**30.0**	**33.9**
Change in public sector debt	-0.7	-2.0	1.2	-0.1	0.2	0.1	1.1	1.7	2.1	0.7	-0.1		
Identified debt-creating flows	**-1.5**	**-1.7**	**-1.4**	**-0.3**	**-0.4**	**0.3**	**0.6**	**1.0**	**1.2**	**0.7**	**-0.1**	**-0.4**	**0.5**
Primary deficit	**1.0**	**1.0**	**0.7**	**1.9**	**1.7**	**2.4**	**2.6**	**3.0**	**3.3**	**3.3**	**2.7**	**1.9**	**2.8**
Revenue and grants	19.6	20.8	22.1	22.0	21.9	21.5	21.2	21.0	20.8	21.3	22.4	**18.3**	**21.3**
of which: grants	1.9	2.5	1.9	1.5	1.3	0.9	0.8	0.6	0.5	0.2	0.0		
Primary (noninterest) expenditure	20.6	21.9	22.9	23.8	23.6	23.8	23.8	24.0	24.1	24.6	25.1	**20.2**	**24.0**
Automatic debt dynamics	**-2.5**	**-2.7**	**-2.2**	**-2.2**	**-2.1**	**-2.1**	**-2.0**	**-2.0**	**-2.1**	**-2.6**	**-2.8**		
Contribution from interest rate/growth differential	-2.3	-1.9	-1.9	-2.2	-2.1	-2.1	-2.0	-2.0	-2.1	-2.5	-2.5		
of which: contribution from average real interest rate	-0.2	0.1	0.0	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.1		
of which: contribution from real GDP growth	-2.1	-2.0	-1.9	-2.0	-2.0	-1.9	-1.9	-1.8	-1.9	-2.3	-2.4		
Contribution from real exchange rate depreciation	-0.2	-0.8	-0.3	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**0.7**	**-0.3**	**2.6**	**0.2**	**0.6**	**-0.2**	**0.5**	**0.7**	**0.9**	**-0.1**	**-0.3**	**0.5**	**0.4**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**23.7**	**23.4**	**23.3**	**23.1**	**23.7**	**24.7**	**26.0**	**30.2**	**31.5**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**107.3**	**106.7**	**106.4**	**107.8**	**111.7**	**117.5**	**124.6**	**141.8**	**140.8**		
Debt service-to-revenue and grants ratio 3/	**2.9**	**4.5**	**4.3**	**4.9**	**5.2**	**5.9**	**6.0**	**6.3**	**6.3**	**7.4**	**10.6**		
Gross financing need 4/	1.3	1.6	1.7	2.9	2.8	3.6	3.9	4.3	4.6	4.8	5.1		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	7.0	6.9	7.0	7.3	7.0	6.7	6.5	6.2	6.0	6.0	6.0	**6.3**	**6.4**
Average nominal interest rate on external debt (in percent)	1.0	1.3	1.2	1.3	1.3	1.3	1.3	1.3	1.2	1.3	1.5	**1.0**	**1.3**
Average real interest rate on domestic debt (in percent)	-0.7	-2.2	-2.1	-2.8	-2.8	-2.9	-2.9	1.8	1.9	1.4	0.8	**-2.3**	**0.0**
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.5	-2.7	-1.0	...	...	...	...	...	...	...	...	**-1.5**	**...**
Inflation rate (GDP deflator, in percent)	1.7	3.5	3.4	2.9	2.9	2.9	3.0	3.0	3.0	3.5	4.1	**3.5**	**3.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	3.3	13.2	11.9	11.8	6.0	7.7	6.6	7.1	6.6	7.4	6.3	**15.8**	**7.1**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	1.8	3.1	-0.4	2.0	1.5	2.3	1.6	1.3	1.2	2.6	2.8	**1.5**	**1.8**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, government-guaranteed debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

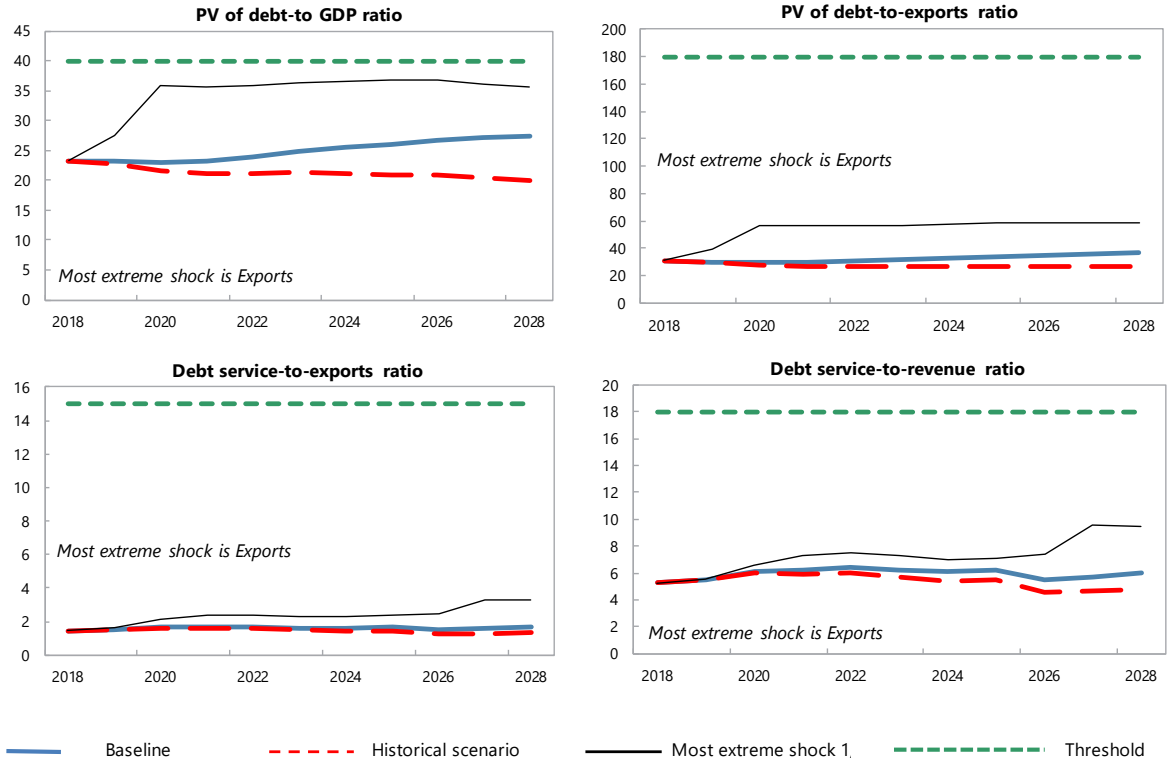
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Cambodia: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2018–2028 1/



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.2%	1.2%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	25	25
Avg. grace period	6	6

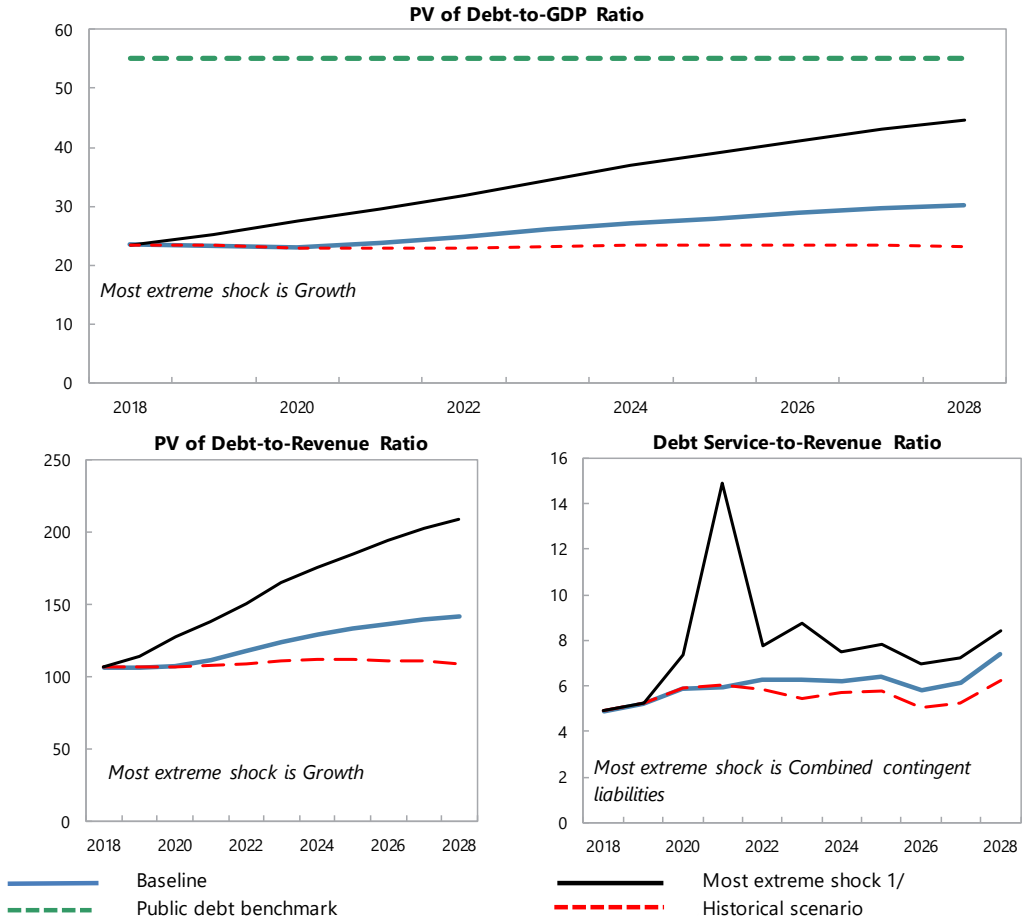
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Cambodia: Indicators of Public Debt Under Alternative Scenarios, 2018–2028 1/



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	93%	83%
Domestic medium and long-term	7%	17%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.2%	2.7%
Avg. maturity (incl. grace period)	25	25
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	1.8%	2.0%
Avg. maturity (incl. grace period)	7	3
Avg. grace period	6	1
Domestic short-term debt		
Avg. real interest rate	0%	3.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Cambodia Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018–2028

(In Percent)

	Projections										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	**23.3**	**23.2**	**23.0**	**23.2**	**23.8**	**24.8**	**25.5**	**26.0**	**26.6**	**27.1**	**27.5**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	23.3	22.6	21.6	21.2	21.0	21.2	21.2	21.0	20.8	20.4	20.0
B. Bound Tests											
B1. Real GDP growth	23.3	24.3	25.1	25.3	26.0	27.0	27.8	28.4	29.1	29.5	30.0
B2. Primary balance	23.3	25.0	26.1	26.4	27.1	28.0	28.6	29.0	29.5	29.8	30.0
B3. Exports	23.3	27.5	35.9	35.6	35.8	36.3	36.6	36.7	36.7	36.1	35.6
B4. Other flows 2/	23.3	25.8	27.5	27.5	28.0	28.8	29.3	29.7	30.1	30.2	30.2
B6. One-time 30 percent nominal depreciation	23.3	29.2	25.8	26.2	27.1	28.3	29.3	30.1	31.0	31.8	32.6
B6. Combination of B1-B5	23.3	29.7	30.5	30.6	31.1	31.9	32.5	33.0	33.2	33.3	33.4
C. Tailored Tests											
C1. Combined contingent liabilities	23.3	30.3	29.8	30.8	31.1	32.0	32.4	32.7	33.1	33.2	33.4
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**	**40**
PV of debt-to-exports ratio											
Baseline	**31.1**	**30.2**	**29.7**	**29.9**	**30.6**	**31.6**	**32.8**	**33.9**	**35.0**	**35.9**	**36.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	31.1	29.5	27.9	27.2	26.9	27.1	27.3	27.3	27.3	27.1	26.8
B. Bound Tests											
B1. Real GDP growth	31.1	30.2	29.7	29.9	30.6	31.6	32.8	33.9	35.0	35.9	36.8
B2. Primary balance	31.1	32.5	33.7	34.0	34.8	35.7	36.9	37.8	38.7	39.5	40.2
B3. Exports	31.1	39.8	56.9	56.2	56.2	56.8	57.8	58.5	59.0	58.7	58.4
B4. Other flows 2/	31.1	33.6	35.5	35.4	35.9	36.7	37.8	38.7	39.5	40.0	40.5
B6. One-time 30 percent nominal depreciation	31.1	30.2	26.4	26.7	27.5	28.7	30.0	31.1	32.3	33.5	34.6
B6. Combination of B1-B5	31.1	37.8	35.9	39.5	40.0	40.9	42.1	43.1	43.8	44.3	44.9
C. Tailored Tests											
C1. Combined contingent liabilities	31.1	39.5	38.6	39.6	39.9	40.8	41.8	42.6	43.4	44.1	44.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**180**
Debt service-to-exports ratio											
Baseline	**1.4**	**1.5**	**1.6**	**1.6**	**1.7**	**1.6**	**1.6**	**1.7**	**1.5**	**1.6**	**1.7**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	1.4	1.5	1.6	1.6	1.6	1.5	1.4	1.5	1.3	1.3	1.3
B. Bound Tests											
B1. Real GDP growth	1.4	1.5	1.6	1.6	1.7	1.6	1.6	1.7	1.5	1.6	1.7
B2. Primary balance	1.4	1.5	1.7	1.8	1.8	1.8	1.7	1.8	1.7	1.9	2.0
B3. Exports	1.4	1.6	2.1	2.4	2.4	2.3	2.3	2.3	2.5	3.2	3.3
B4. Other flows 2/	1.4	1.5	1.7	1.7	1.8	1.7	1.7	1.8	1.7	2.0	2.0
B6. One-time 30 percent nominal depreciation	1.4	1.5	1.6	1.6	1.6	1.6	1.6	1.6	1.4	1.4	1.5
B6. Combination of B1-B5	1.4	1.6	1.9	1.9	2.0	1.9	1.9	1.9	2.1	2.2	2.3
C. Tailored Tests											
C1. Combined contingent liabilities	1.4	1.5	1.9	1.9	2.0	1.9	1.9	1.9	1.8	1.8	1.9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**
Debt service-to-revenue ratio											
Baseline	**5.3**	**5.5**	**6.2**	**6.2**	**6.4**	**6.3**	**6.1**	**6.2**	**5.5**	**5.7**	**6.0**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	5.3	5.5	6.0	5.9	6.0	5.7	5.4	5.5	4.6	4.6	4.7
B. Bound Tests											
B1. Real GDP growth	5.3	5.8	6.7	6.8	7.0	6.8	6.6	6.8	6.0	6.3	6.6
B2. Primary balance	5.3	5.5	6.4	6.7	6.9	6.8	6.6	6.7	6.3	6.7	7.0
B3. Exports	5.3	5.5	6.6	7.3	7.5	7.3	7.0	7.1	7.4	9.5	9.5
B4. Other flows 2/	5.3	5.5	6.4	6.6	6.8	6.6	6.4	6.5	6.4	7.1	7.2
B6. One-time 30 percent nominal depreciation	5.3	7.0	7.8	7.5	7.8	7.6	7.4	7.6	6.7	6.3	6.7
B6. Combination of B1-B5	5.3	5.9	7.2	7.3	7.5	7.3	7.0	7.2	7.5	7.8	8.0
C. Tailored Tests											
C1. Combined contingent liabilities	5.3	5.5	7.3	7.3	7.6	7.4	7.2	7.3	6.5	6.7	6.9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**	**18**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Includes official and private transfers and FDI.

Table 4. Cambodia Sensitivity Analysis for Key Indicators of Public Debt, 2018–2028

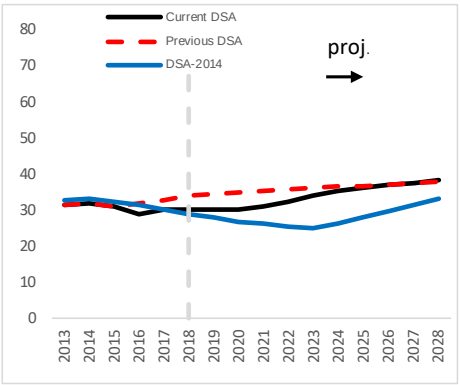
(In percent)

	Projections										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	**23.4**	**23.3**	**23.1**	**23.7**	**24.7**	**26.0**	**27.0**	**27.9**	**28.9**	**29.6**	**30.2**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	23	24	23	23	23	23	23	23	23	23	23
B. Bound Tests											
B1. Real GDP growth	23	25	27	29	32	34	37	39	41	43	45
B2. Primary balance	23	26	27	27	28	29	30	31	32	32	33
B3. Exports	23	28	36	36	37	38	38	39	39	39	38
B4. Other flows 2/	23	26	28	28	29	30	31	32	32	33	33
B6. One-time 30 percent nominal depreciation	23	28	26	25	24	23	22	22	21	21	20
B6. Combination of B1-B5	23	24	24	23	24	26	27	27	28	29	30
C. Tailored Tests											
C1. Combined contingent liabilities	23	32	32	32	32	33	34	35	35	36	36
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of Debt-to-Revenue Ratio											
Baseline	**106.7**	**106.4**	**107.8**	**111.7**	**117.5**	**124.6**	**129.4**	**133.2**	**137.1**	**140.0**	**141.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	107	107	107	108	109	111	112	112	112	111	109
B. Bound Tests											
B1. Real GDP growth	107	115	128	138	151	165	176	185	195	203	209
B2. Primary balance	107	117	126	129	134	141	145	148	151	153	154
B3. Exports	107	127	168	170	174	180	183	184	185	183	180
B4. Other flows 2/	107	118	129	132	137	144	148	151	154	155	155
B6. One-time 30 percent nominal depreciation	107	130	122	117	113	111	107	103	100	97	93
B6. Combination of B1-B5	107	111	111	110	115	122	127	131	135	138	139
C. Tailored Tests											
C1. Combined contingent liabilities	107	148	148	149	154	160	163	166	168	170	170
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	**4.9**	**5.2**	**5.9**	**6.0**	**6.3**	**6.3**	**6.2**	**6.4**	**5.8**	**6.1**	**7.4**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	5	5	6	6	6	5	6	6	5	5	6
B. Bound Tests											
B1. Real GDP growth	5	5	7	8	9	9	9	10	9	10	12
B2. Primary balance	5	5	6	9	8	7	7	7	7	7	8
B3. Exports	5	5	6	7	7	7	7	7	8	10	11
B4. Other flows 2/	5	5	6	6	7	7	6	7	7	7	9
B6. One-time 30 percent nominal depreciation	5	6	7	7	7	7	7	7	6	6	7
B6. Combination of B1-B5	5	5	6	6	6	6	6	6	6	6	7
C. Tailored Tests											
C1. Combined contingent liabilities	5	5	7	15	8	9	7	8	7	7	8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

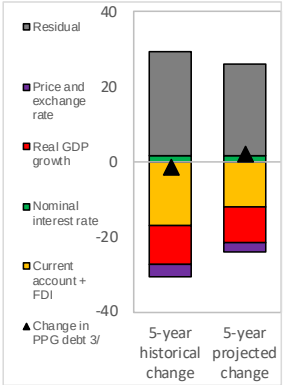
Sources: Country authorities; and staff estimates and projections.
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
2/ Includes official and private transfers and FDI.

Figure 3. Cambodia: Drivers of Debt Dynamics – Baseline Scenario External Debt

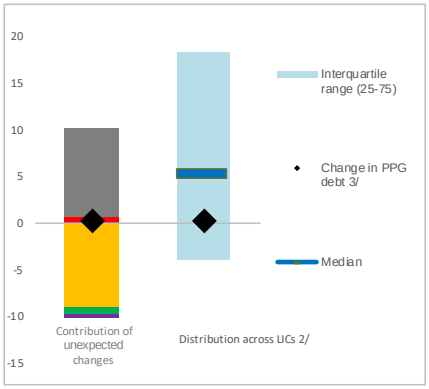
Gross Nominal PPG External Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)

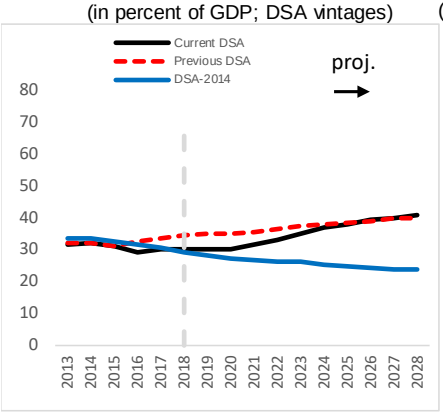


Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)

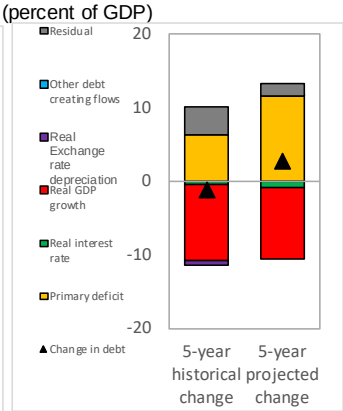


Public debt

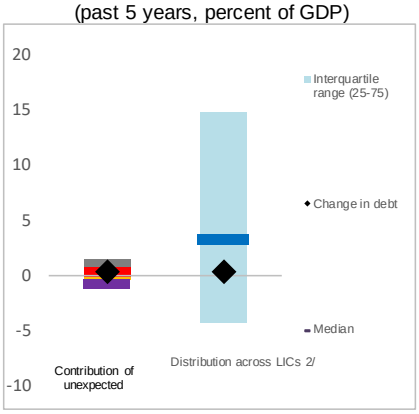
Gross Nominal Public Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)



Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)



Sources: Country authorities; and staff estimates and projections.

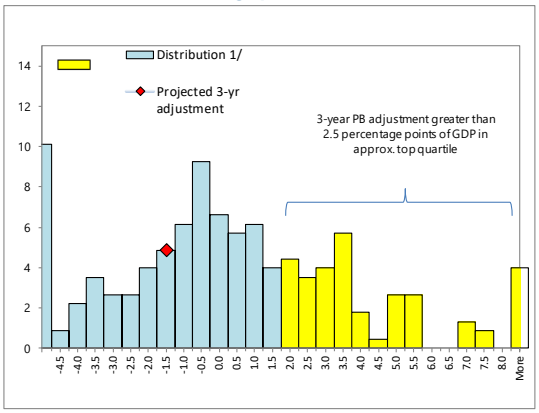
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

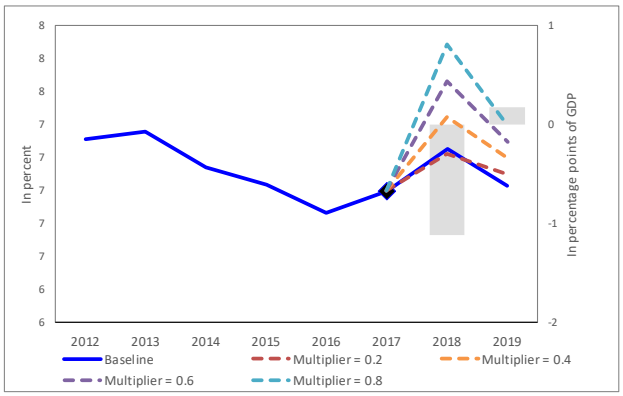
Figure 4. Cambodia: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



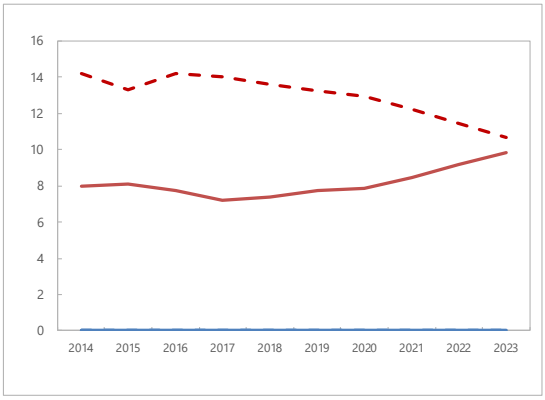
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



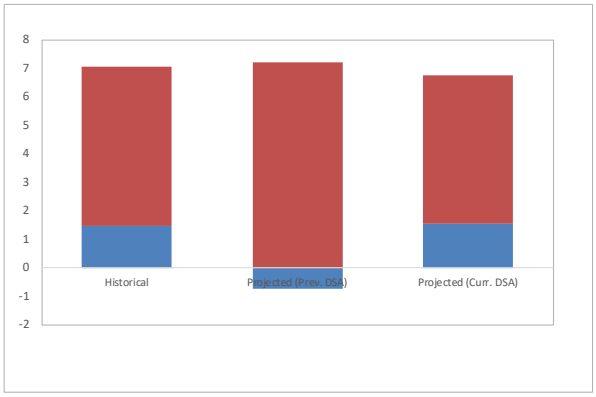
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Gov. Invest. - Prev. DSA
Gov. Invest. - Current DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Current DSA

Contribution to Real GDP growth
(percent, 5-year average)



Contribution of other factors
Contribution of government capital

Sources: Country authorities; and staff estimates and projections.